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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

GRAPH AND WEB ANALYTICS - 23DSB3511

A.Y. 2026 - 2027

TITLE OF THE PROJECT	GRAPH-BASED CLICKSTREAM ANALYSIS FOR WEB NAVIGATION AND USER BEHAVIOR	
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Abstract

The project titled Graph-Based Clickstream Analysis for Web Navigation and User Behavior is aimed at analyzing users' navigation through a website as well as identifying relations between the website pages. The clickstream data of a website includes users' activities, pages they have visited, the time when those events have happened, sessions, traffic sources, and various events. At that, no information on users' navigation is present in that data.

In this project, clickstream data is preprocessed in Google Colab utilizing Python. In particular, the dataset is checked for its structure, missing data, duplicated data, and data types. Then, the event time is transformed into a datetime and the activities of users are sorted by their chronological order. It allows recognizing the order of the webpages visited by the users.

The identified navigation sequences are further transformed into a graph. In such a graph, the webpages are nodes, while the movement from one page to another is directed edges. The frequency of page-to-page transition is the edge weight. This representation helps to find the relations between webpages and users' navigation patterns.

The graph of navigation is further analyzed using . In particular, Dijkstra's algorithm is applied to identify the right navigation paths between the webpages. Also, such measures of importance and connectivity of webpages as Degree Centrality, Closeness Centrality, Betweenness Centrality, and Eigenvector Centrality are calculated. Furthermore, PageRank is used to recognize the structurally important webpages, and Community Detection – to identify groups of webpages which are connected.

Besides the graph analysis, web analytics including information on users, sessions, traffic sources, devices, countries, keywords, conversions, and session duration are analyzed. These results are visualized through the Streamlit dashboard where users can check the website navigation graph, metrics of the graph, navigation paths, communities, and results of web analytics.

Signature of the Guide HOD